

Trainings ▾

General Information

Front Gear Train engines are equipped with:

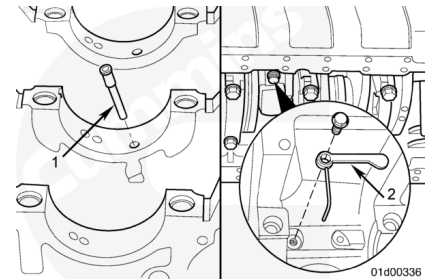
1. Saddle Jet Piston Cooling Nozzles

Rear Gear Train engines are equipped with:

2. J-jet Piston Cooling Nozzles

Saddle jet piston cooling nozzles are located in the main bearing saddle on the block side. Oil is supplied from the main bearing.

J-jet piston cooling nozzles are located in between the main bearing saddles on the exhaust side of the engine. Oil is supplied from an oil gallery in the block on the exhaust side of the engine.

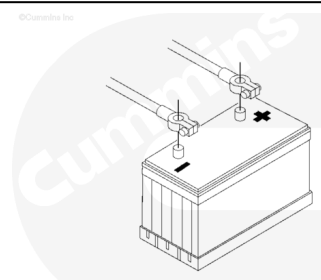


Preparatory Steps

⚠ WARNING ⚠

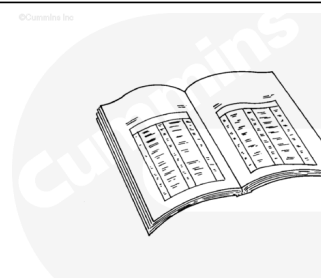
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>)
- Remove the oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>)
- If equipped, remove the block stiffener plate. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html>).

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

Note : It is not necessary to remove the crankshaft if removing J-jet piston cooling nozzles.

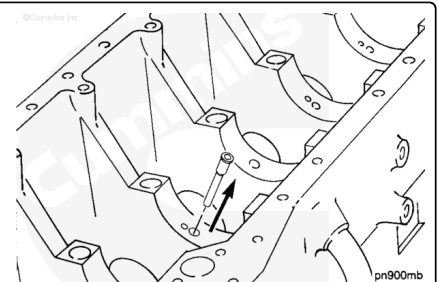
- Remove the crankshaft. Refer to Procedure 001-016 (</qs3/pubsys2/xml/en/procedures/40/40-001-016-tr.html>).

Remove

Saddle Jet Piston Cooling Nozzle

Remove the piston cooling nozzles by pressing from the top with an appropriate size punch.

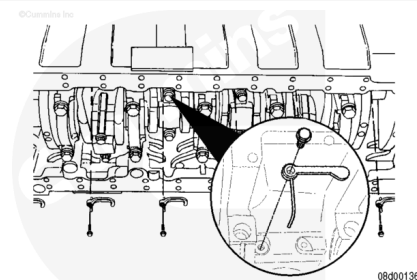
Note : Do not reuse the saddle jet piston cooling nozzles once removed.



J-Jet Piston Cooling Nozzle

Rotate the crankshaft to various positions to access each piston cooling nozzle.

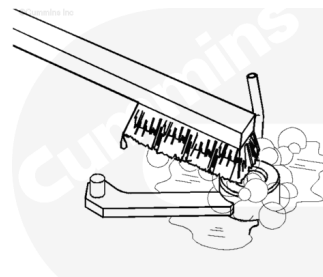
Remove the piston cooling nozzles.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : Do not reuse the saddle jet piston cooling nozzles or plugs once removed.

Clean the J-jet piston cooling nozzle and oil passage in the block using solvent. Dry with compressed air.

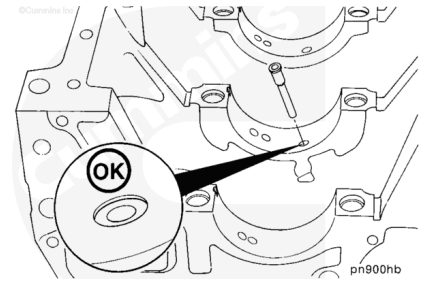
Inspect the capscrew and J-jet piston cooling nozzle for damage.

Install

Saddle Jet Piston Cooling Nozzle

⚠ CAUTION ⚠

Do not use a hammer to install the piston cooling nozzles. Use hand pressure only. Using a hammer can cause component damage.



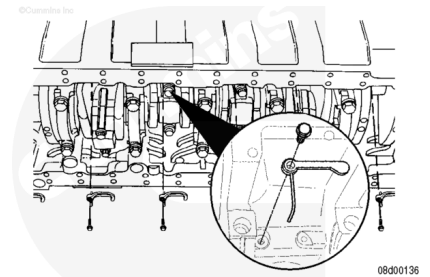
Push the piston cooling nozzle into place by hand.

Use a flat punch to push the nozzle into the recess.

J-Jet Piston Cooling Nozzle

⚠ CAUTION ⚠

Slight bending of the piston cooling nozzles can result in severe engine damage. Replace piston cooling nozzle if it is bent or damaged during disassembly or assembly.



Install the piston cooling nozzle one cylinder at a time rotating the crankshaft as necessary for access.

Use a long extension to guide the capscrew and/or piston cooling nozzle into place.

Note : The locator pin on the J-jet piston cooling nozzle must engage the locating hole in the block for proper alignment.

Tighten the capscrew.

J-Jet Capscrew	15 n.m	[133 in-lb]
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Finishing Steps

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



- If removed, install the crankshaft. Refer to Procedure 001-016 (/qs3/pubsys2/xml/en/procedures/40/40-001-016-tr.html)
- If removed, install the block stiffener plate. Refer to Procedure 001-089

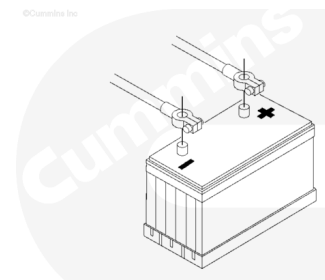
(/qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html)

- Install the lubricating oil suction tube. Refer to Procedure 007-035 (/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)
- Install the oil pan and oil pan gasket. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/40/40-007-037.html).

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries
- Operate the engine and check for leaks and proper oil pressure.



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